

Evaluating and Equivalent Expressions*Independent Set 3.06* **ANSWER KEY***Six questions, on your own, in silence. Show the step that does the work.*

1. Find the value. $\frac{(-5)^4}{(-5)^0 \cdot (-5)^2}$

$$(-5)^2 = 25$$

$$4 - 0 - 2 = 2.$$

2. Find the value. $\left(\frac{2}{3}\right)^{-2} \cdot \left(\frac{2}{3}\right)^3$

$$\frac{2}{3}$$

$$-2 + 3 = 1.$$

3. Write as a power of 2. 16^3

$$2^{12}$$

$$16 = 2^4.$$

4. Write as a power of 3. $27^2 \cdot 9$

$$3^8$$

$$3^6 \cdot 3^2.$$

5. Are 4^{-2} and 2^{-4} equivalent? Show why.

Yes — both are $\frac{1}{16}$.

$$4^{-2} = (2^2)^{-2} = 2^{-4}.$$

6. Find the value. $(3^{-1})^2 \cdot 3^5$

$$3^3 = 27$$

$$-2 + 5 = 3.$$